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RESEARCH DATA MANAGEMENT AND THE FAIR PRINCIPLES

Advanced microscopy: valorization and collaboration with industry, Paula Oset, 18/09/2025
https://osf.io/7gxt4

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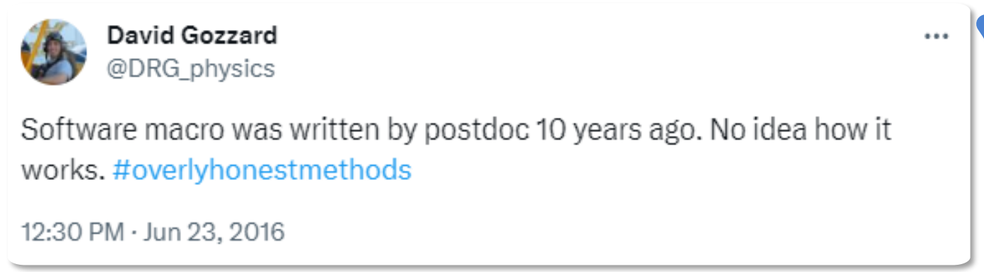
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OUTLINE

- What is Research Data Management?
- Data Management Plans
- The FAIR principles and data repositories

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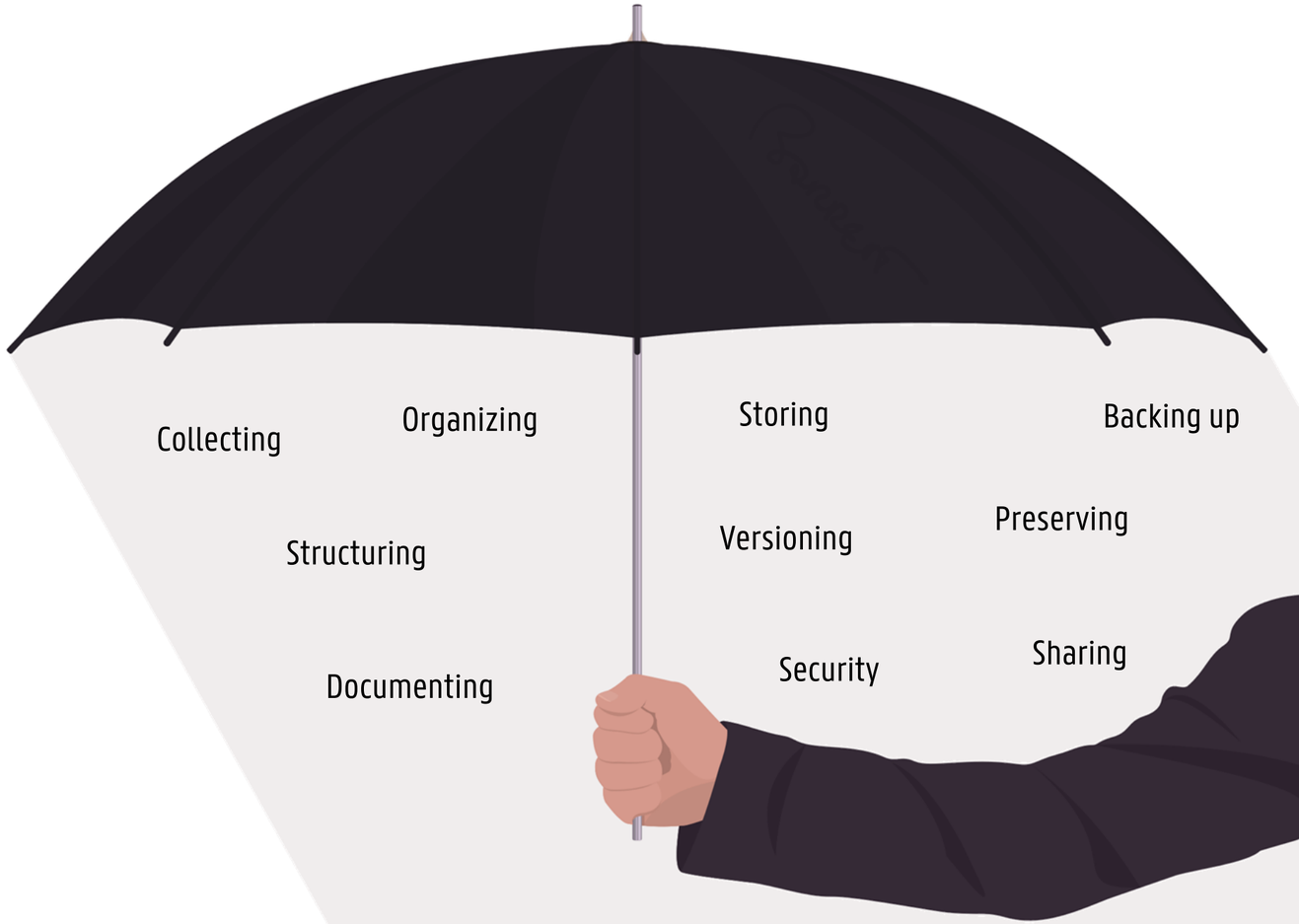
WHAT COULD GO WRONG?

- 3 years into the project, data is lost
- Data exists but can't be reused (no documentation, corrupted, outdated format…)
- Disagreement between project partners on exploitation of results
- Can't keep track of your own files

WHAT IS RESEARCH DATA MANAGEMENT (RDM)?

- Actions to **take care of data during the course of research and beyond**.
- An integral part of research.

"the compilation of many small practices that make your data easier to understand, less likely to be lost, and more likely to be usable during a project or ten years later" (Briney 2015)



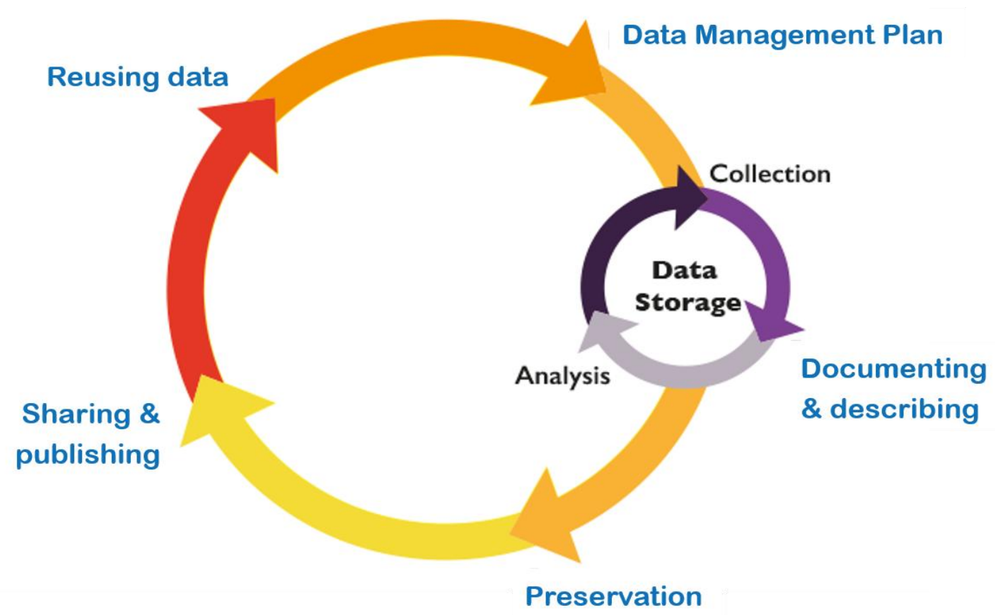
Adapted from 'RDM: An Overview' by Research Support Team, IT Services (University of Oxford), licensed under CC-BY-NC-SA 4.0 and OpenClipart CC BY

DATA MANAGEMENT PLANS (DMP)

- Outlines how research data will be handled **during** & **after** a project.
- Increasingly required by research funders/institutions

  -> as a **first step** towards good research data management (RDM)

  -> requires times up-front, saves time & problems in the long run
- DMPonline.be
- DMP examples
  - By UGent researchers, curated by UGent data stewards



Adapted from DCC

DMP SECTIONS



Data summary & description

Purpose, origin, type, sensitiveness, format & volume



Documentation & metadata

Information to keep data findable, understandable, reusable



Roles & responsibilities



Ethical & legal issues

Intellectual property, GDPR, etc.



Data sharing and reuse

Data repositories, access conditions, licenses or agreements



Costs & resources



Data storage & security

Security measures, back-up, data transfers & access control



Long-term preservation

What to preserve & where, for how long, access restrictions

RDM CHALLENGES IN ADVANCED MICROSCOPY

- High volume and complexity
- Proprietary formats (vendor-specific)
- Lack of harmonized (meta)data standards
- Fragmented software environments

- Collaboration with industry:
  - IP, ownership of results

- Personal data?
  - GDPR

DMP SECTIONS



Data summary & description

Purpose, origin, type, sensitiveness, format & volume

Documentation & metadata

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Roles & responsibilities

Ethical & legal issues

Intellectual property, GDPR, etc.



Long-term preservation

File types, data archiving, for how long, access restrictions



Costs & resources

Describe the data that will be used and/or generated during the research project

Different data types require different management approaches

First step to identify needs and design data management strategy

Data storage & security

Security measures, back-up, data transfers & access control

Data sharing and reuse

Data repositories, access conditions, licenses or agreements

DMP SECTIONS

Data summary & description

Purpose, origin, type, sensitiveness, format & volume

Documentation & metadata

Information to keep data findable,

Roles & responsibilities

Ethical & legal issues

Intellectual property, GDPR, etc.

What to preserve & where

Costs & resources

Data storage & security

Security measures, back-up, data transfers & access control

Data repositories, access conditions, licenses or agreements

- Data “ownership” & access rights
- IP protection
- Licensing & reuse
- Personal data? GDPR
- **Legal support!**
  - Negotiate terms for data sharing early on, document them carefully in the DMP, and formalize them in appropriate legal agreements.

DMP SECTIONS

Data summary & description

Purpose, origin, type, sensitiveness, format & volume

Documentation & metadata

Information to keep data findable, understandable, reusable

Roles & responsibilities

Ethical & legal issues

Intellectual property, GDPR, etc.

Long-term preservation

What to preserve & where, for how long, access restrictions

Costs & resources



Data storage & security

Security measures, back-up, data transfers & access control

- Risks vs. security
- Data storage & backup procedures
- Access control
- Other (physical) security measures

THE DO'S AND DON'TS OF DATA SECURITY

Avoid local storage or own IT resources

Avoid storing confidential information in third-party cloud services

Do not send data by email

Use institutional storage

Backup regularly

Configure access permissions

Encrypt confidential data

Use safe file transfer alternatives

Keep your system updated

Use strong passwords

Protect your hardware

Control physical access to confidential data

When needed, destroy information in a safe way

- Shred paper
- Erasing software
- Destroy hardware

More: https://helpdesk.ugent.be/security/en/
Knowledge clip: Data security

RECOMMENDED STORAGE OPTIONS OFFERED BY UGENT



- **SharePoint sites:** cloud based storage offered by UGent DICT
- Large **capacity** but individual file size <250GB
- Easy to **synchronize** to local: accessible anywhere via browser or from file explorer (also offline)
- Share with external users
- Give access to (at least) your promotor
  - (vs. OneDrive) Avoid data is lost when/if your account is suspended
- Automatic **back-up and version control**
- **RDM workspace template** (How to request)
- More info & demo recording

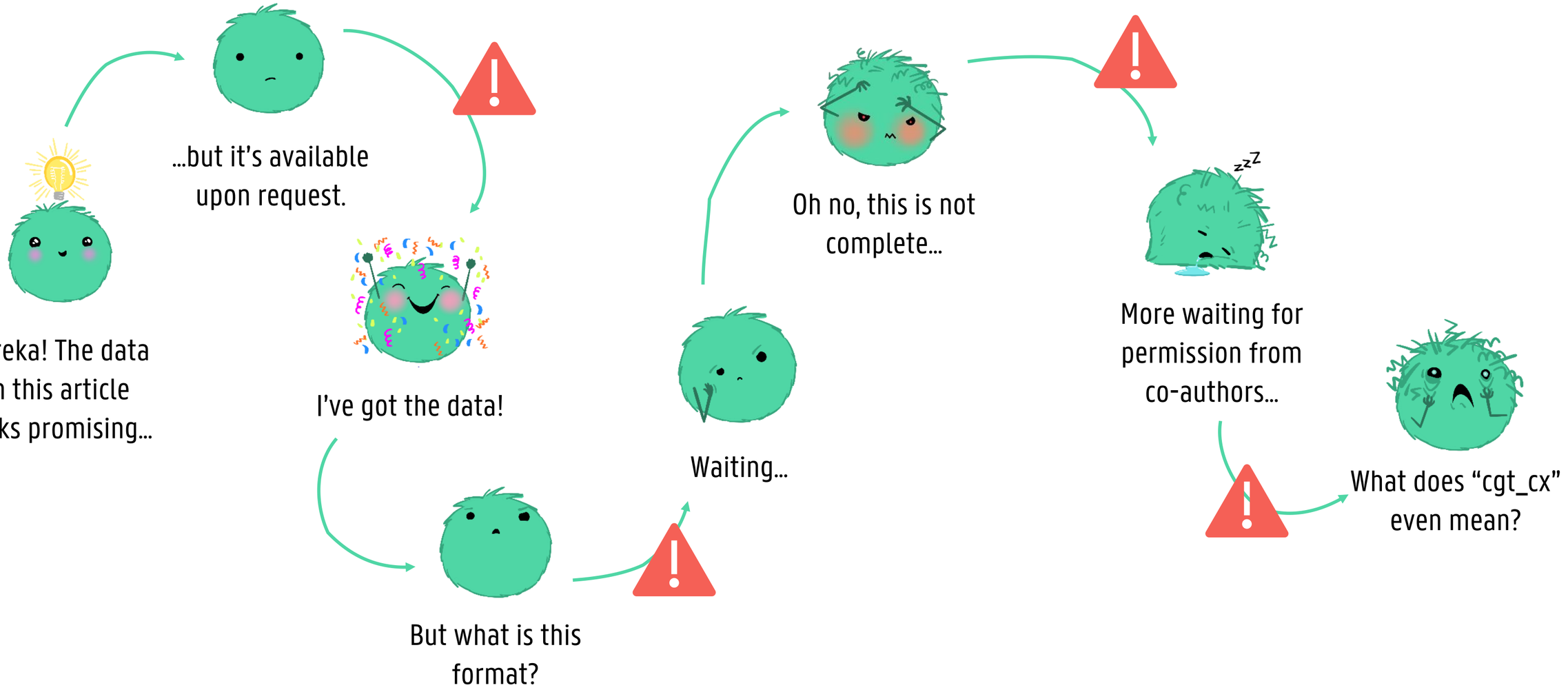


- **Shares:** network storage offered by UGent DICT. Server in S10
- Also known as *vakgroep* shares, project shares, ACL shares, UGent central disk space…
- Large capacity, no file size limit
- Need to be connected to UGent network: work on campus or with VPN
- No offline work
- Automatically backed-up: easy to recover files
- How to request
- More info & demo recording

Other (e.g. department storage)?
Get informed!

THE FAIR PRINCIPLES

OUR WAY TO SHARE & REUSE DATA COULD BE MORE EFFICIENT…



Artwork by Allison Horst

WHY WERE THE FAIR PRINCIPLES CREATED?

- Researchers need better infrastructure to share & reuse data
- Not just about humans: machines need to find, understand and use data too (automatically)
- The FAIR principles are guidelines to make data easier to find, share and use by both humans & machines



The Turing Way Community, & Scriberia. (2020, March 3). Illustrations from the Turing Way book dashes. Zenodo. http://doi.org/10.5281/zenodo.3695300, CCBY

FAIR DATA PRINCIPLES

Can people (and machines) easily find my data?

Once found, is it clear who can access it and how?

Can it be easily integrated with other data? Is my data in common formats and uses terms everyone understands?

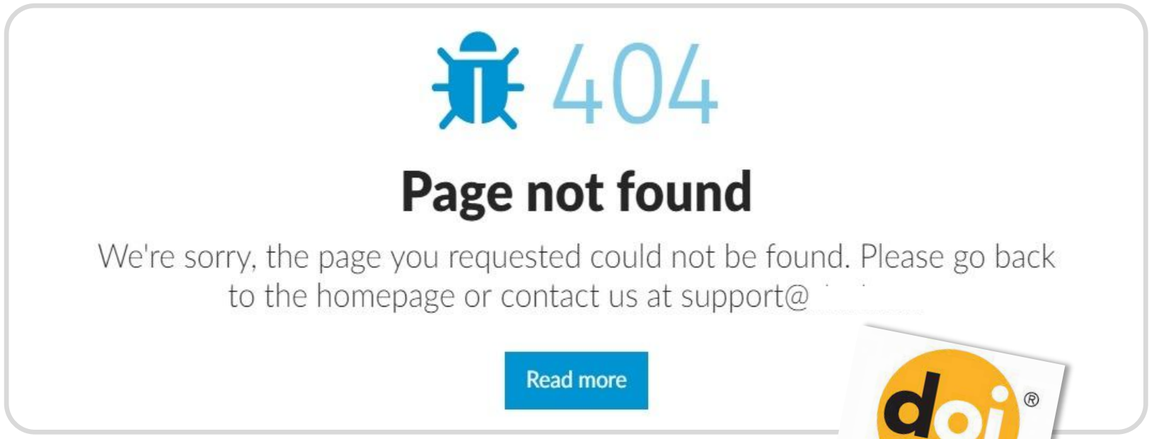
Can someone else understand and use my data? Did I provide clear descriptions, instructions, and license for reuse?

FINDABLE	Persistent identifiers (PIDs)	Rich metadata, includes PID	Machine readable metadata
ACCESSIBLE	Authentication & authorization procedures	Data access via PID & standard comm. protocol	Metadata is always available
INTEROPERABLE	Metadata standards	Vocabularies & ontologies	Linked metadata
REUSABLE	Metadata includes license	Metadata includes provenance	(Meta)data community standards

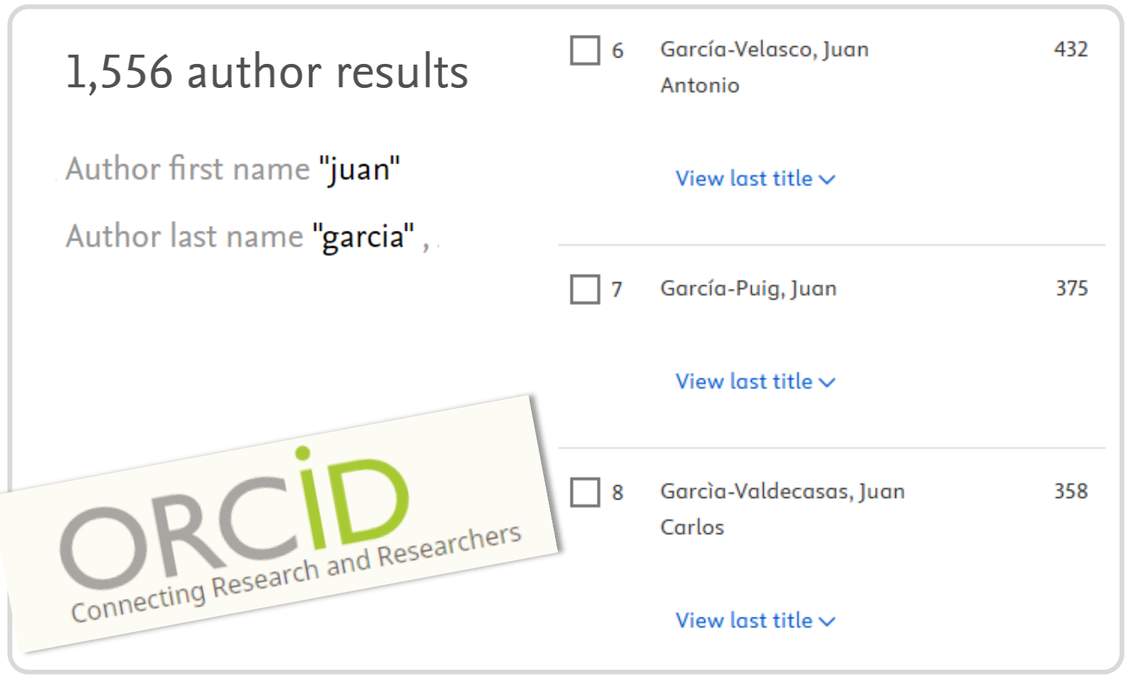
PERSISTENT IDENTIFIERS

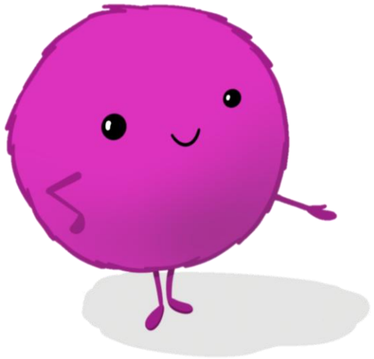
FINDABLE	Persistent identifiers (PIDs)	Rich metadata, includes PID	Machine readable metadata
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ISSUES WITH DATA FINDABILITY AND ACCESSIBILITY









Persistent identifiers:

Permanent and **globally unique** reference to a digital artifact

CROSS-REFERENCE OUTPUTS & TRACK CONTRIBUTIONS

Article

ELSEVIER

Biosystems Engineering
Volume 234, October 2023, Pages 125-139



Research Paper

Sim2real flower detection towards automated Calendula harvesting

Wout Vierbergen [a b], Axel Willekens [a c], Donald Dekeyser [a], Simon Cool [a], Francis wyffels [c]

Show more

+ Add to Mendeley Share Cite

Article DOI

https://doi.org/10.1016/j.biosystemseng.2023.08.016 ↗ Get rights and content ↗

Results

3.1. Data collection and generation

In this study, different datasets were collected and generated to create the proposed pipeline to generate synthetic agricultural data. Besides, a test set of real images was collected to evaluate the sim-to-real transition of the detection model. This section lists the results of the data collection and generation. All collected data, the 3D models of Calendula plants, and the generated [illegible]et are published on Zenodo under CC-BY licence (Vierbergen et al., 2022).

Dataset link

References

Vierbergen et al., 2022 W. Vierbergen, A. Willekens, D. Dekeyser, S. Cool, F. wyffels

Sim2real flower detection towards automated calendula harvesting

Type: Dataset

https://doi.org/10.5281/zenodo.6945367 ↗ (2022)

Dataset DOI

zenodo

Published September 13, 2023 | Version v1 Dataset Open

Sim2real flower detection towards automated Calendula harvesting

Vierbergen, Wout[1]; Axel Willekens[1]; Donald Dekeyser[1]; Simon Cool[1]; Francis wyffels[2]

ORCIDs

Show affiliations

This dataset serves as supplementary material for the research paper titled '*Sim2real flower detection towards automated Calendula harvesting*', which was published in the October 2023 issue of Biosystems Engineering. Within this upload, you will find a collection of both authentic and computer-generated images featuring Calendula (*Calendula officinalis L.*) flowers. Additionally, we have included the resources and original data utilized in generating the synthetic images. This dataset proves instrumental in demonstrating the successful transference of a deep neural network from simulation to real-world applications.

For more comprehensive details regarding this dataset and its associated metadata, we invite you to consult the published article in Biosystems Engineering, available at the following link:

https://doi.org/10.1016/j.biosystemseng.2023.08.016.

Article DOI

Related works **Is supplement to**

Journal article: 10.1016/j.biosystemseng.2023.08.016 (DOI)

Article DOI

Details

DOI

DOI 10.5281/zenodo.6945367

Dataset DOI

Resource type
Dataset

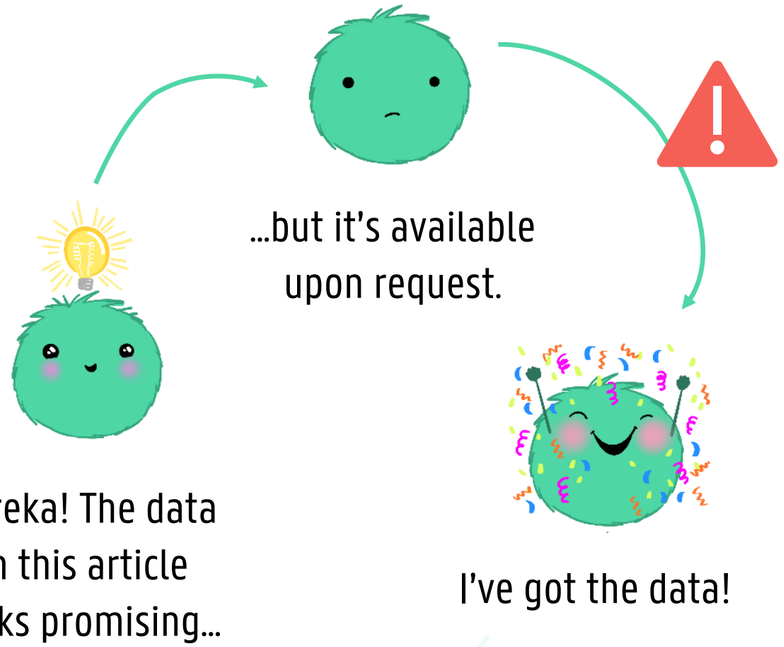
Publisher
Zenodo

Dataset

Relations

Details

WHAT ABOUT OUR RESEARCHER?



- If the data was deposited in a data repository and had a DOI, our researcher could have found and accessed it without the need to contact the authors.
- Data would have been preserved in a safe infrastructure (i.e. data repository), where it is more likely to remain accessible in the long term.

Artwork by Allison Horst

METADATA, DOCUMENTATION & STANDARDS

FINDABLE	Persistent identifiers (PIDs)	Rich metadata, includes PID	Machine readable metadata
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INTEROPERABLE	Metadata standards	Vocabularies & ontologies	Linked metadata
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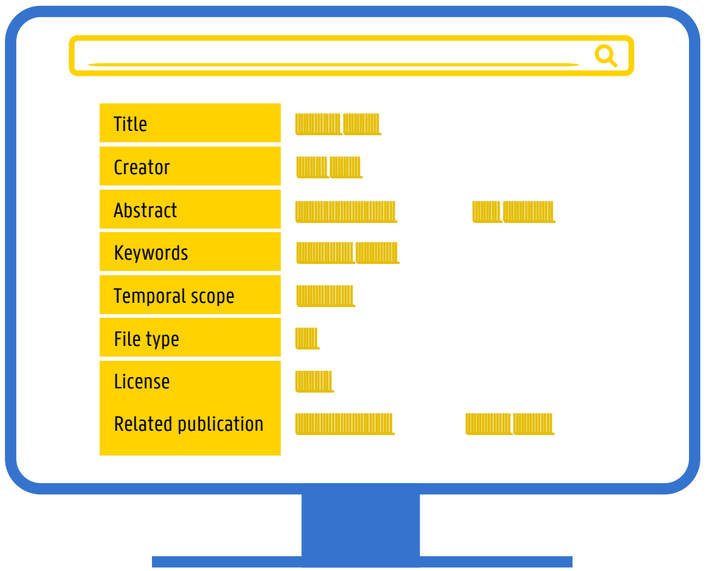
DOCUMENTATION VS. METADATA

WHAT WHEN WHERE HOW WHO WHY

DOCUMENTATION: any contextual information that allows data to be understandable and reusable.

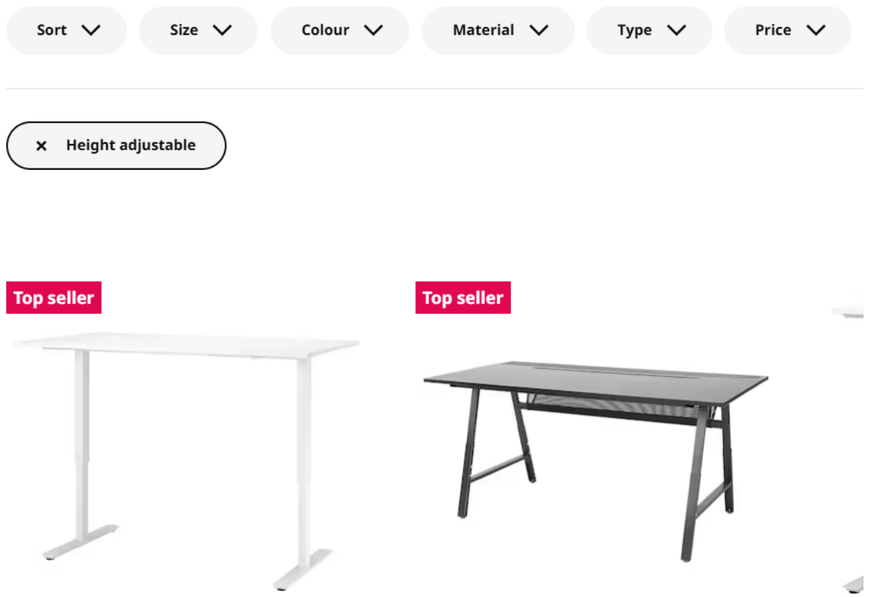
The dataset...

METADATA: information about data, in a **highly structured** way, using a set of defined fields to be **machine readable**.

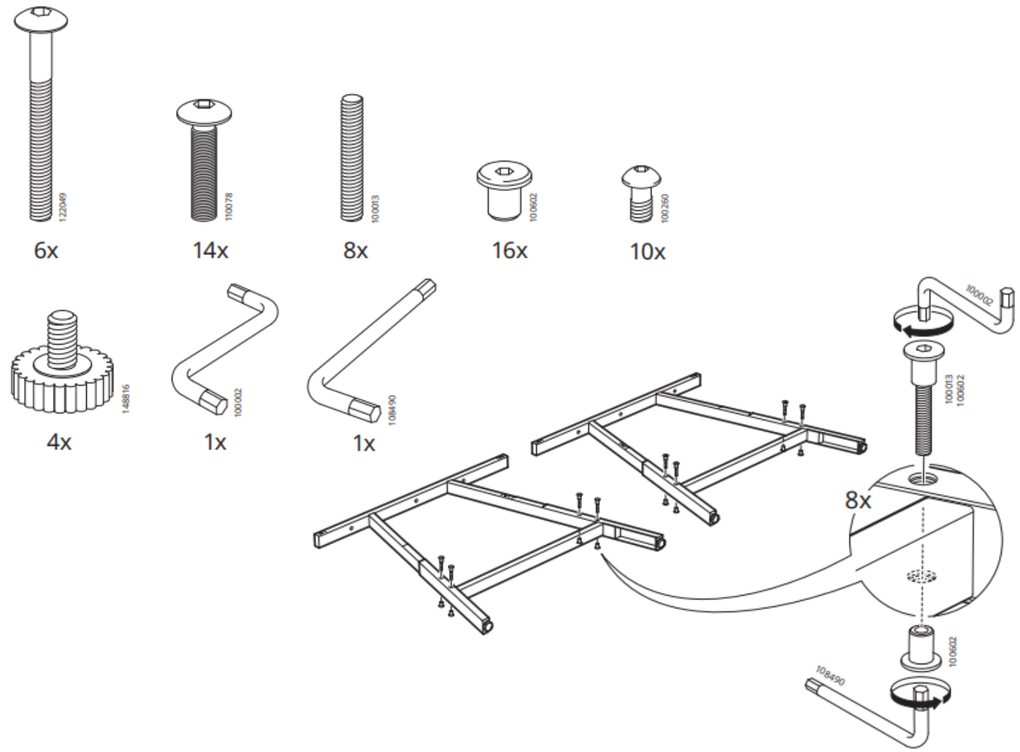


DOCUMENTATION VS. METADATA

- Metadata allows others:
  - To find data
  - To assess whether it is useful

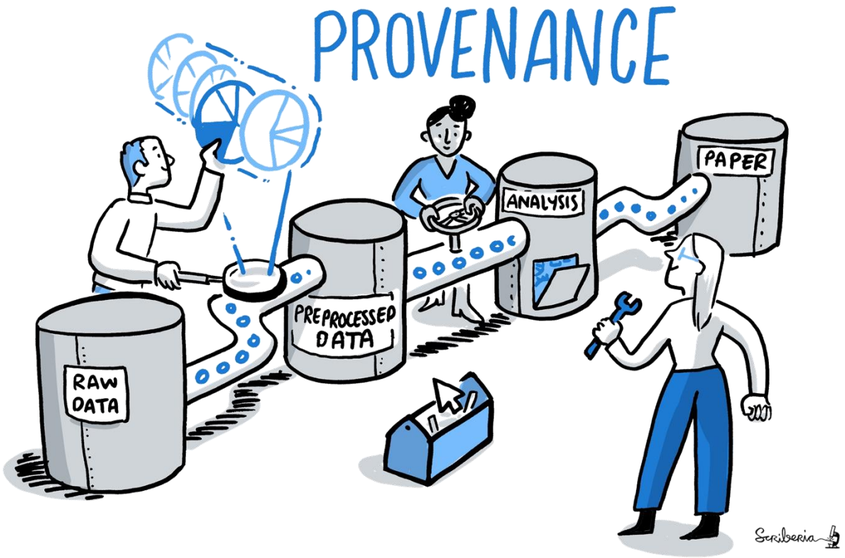


- Documentation allows others:
  - To understand how data was created & processed, and how to (re)use it



DOCUMENTATION

All the contextual information about data that enables others to understand, reuse, and reproduce it.



Sample preparation

Sample type, preparation steps, chemicals / reagents, dates, operators, environmental conditions

Image acquisition

Instrument model, settings (e.g., pixel size, magnification), date/time, scan ID, output format, file naming

Reconstruction / alignment

Software/tool/algorithm used, reconstruction method, parameters, logs, manual interventions.

Processing

What was done, with what tool, settings, version history, and whether raw data was preserved.

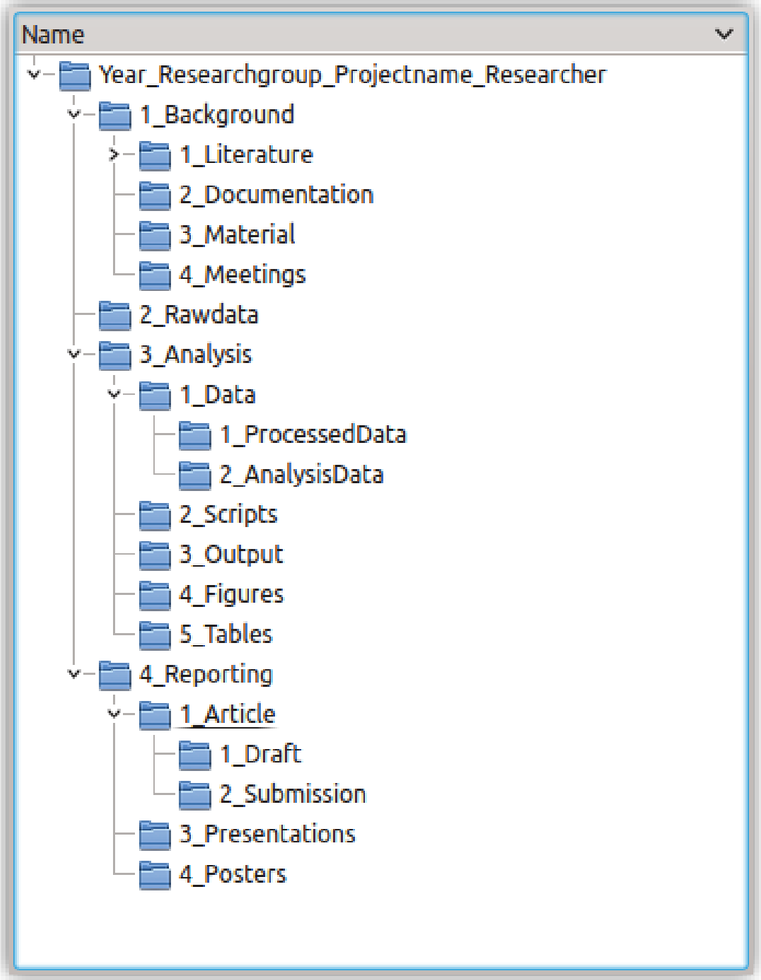
Annotation / segmentation

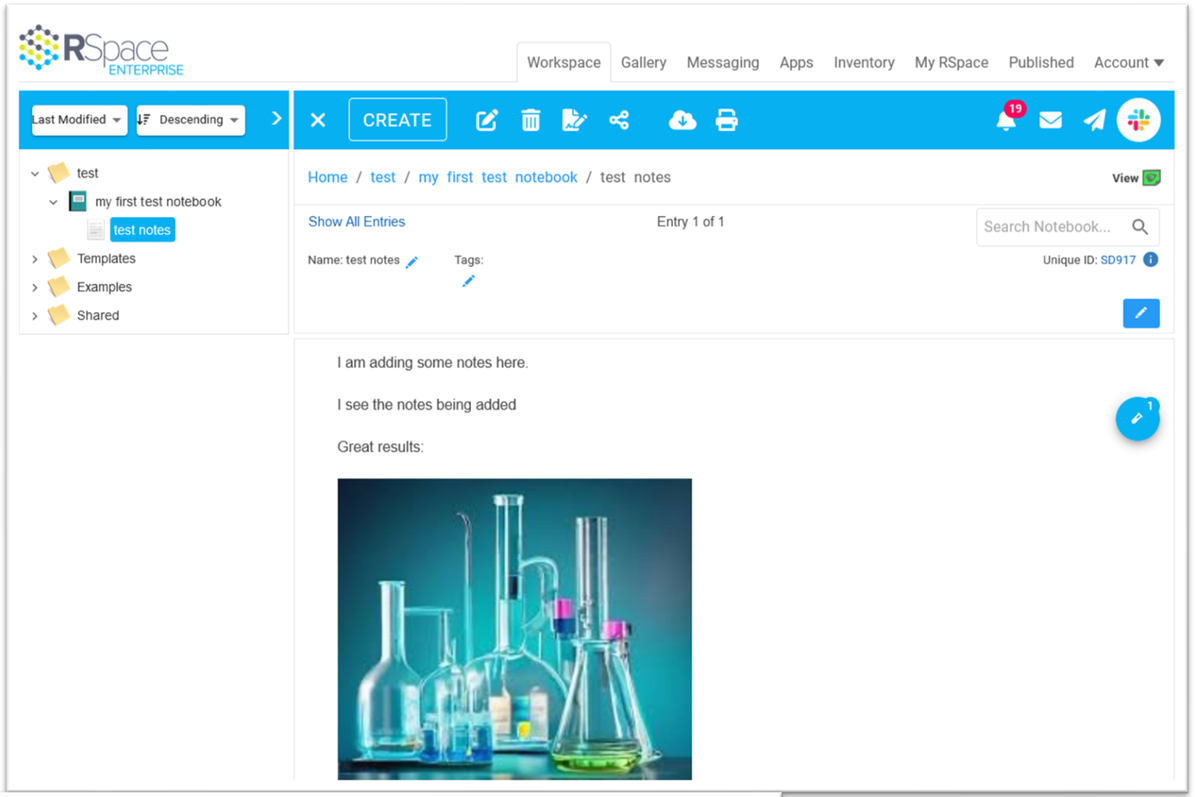
Method used, software version, thresholds or model info, labelling rules, uncertainty notes

Quantitative analysis

Variable definitions & units, software, assumptions, linkage to raw data.

EXAMPLES OF DOCUMENTATION AT DIFFERENT LEVELS

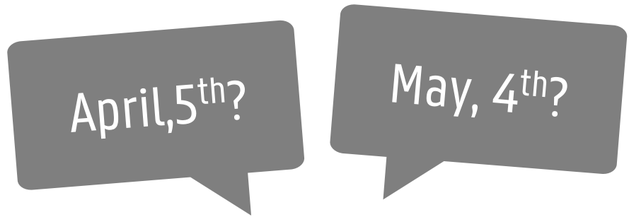




codebook_data columns and labels

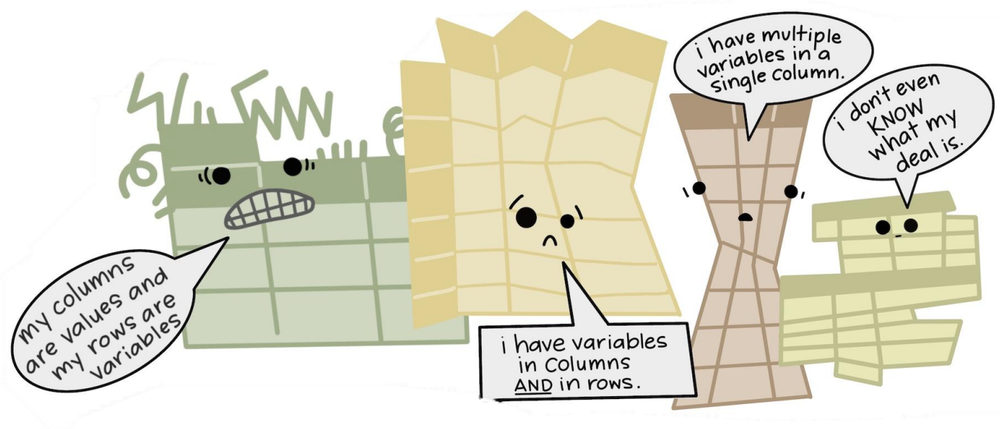
name	label	value_labels
All	All	All
O5R	Will not probe deeply into a subject.	6. Very Inaccurate, 5. Moderately Inaccurate, 4. Slightly Inaccurate, 3. Slightly Accurate, 2. Moderately Accurate, 1. Very Accurate
gender	gender	1. male, 2. female

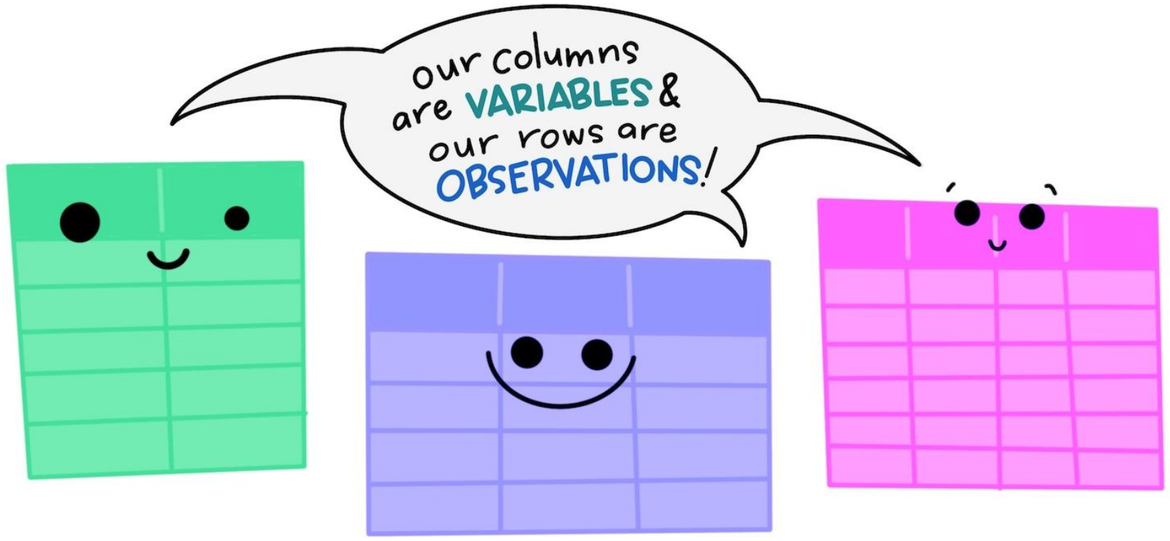
COMMON ISSUES FOR DATA REUSE



05/04/85

ISO standard (YYYYMMDD or YYYY-MM-DD)
1985-04-05





STANDARDS & CONTROLLED VOCABULARIES

- **Standards**: Rules that govern how (meta)data is structured, formatted, shared, and exchanged across systems.
- **Controlled vocabularies**: Define agreed-upon terms to ensure consistency in describing variables, values, units, etc.
- Why?



Unambiguity
Everyone uses the same terms for the same concepts



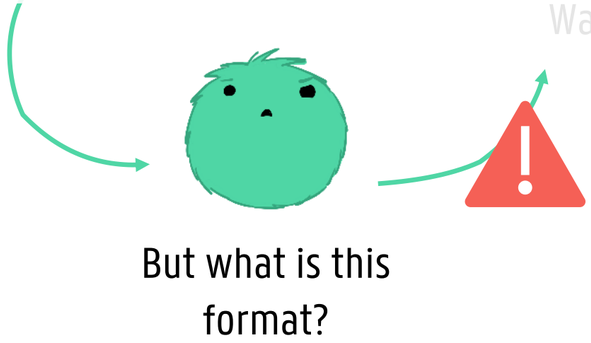
Interoperability
Data from different researchers and disciplines can be easily integrated

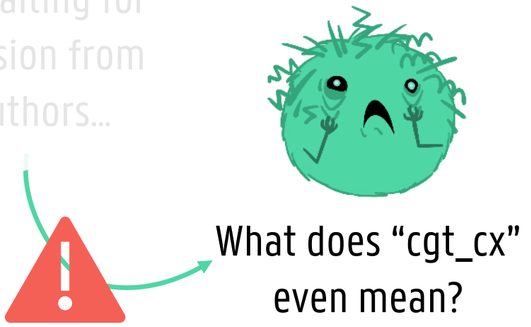


Machine actionability
Computers can process data more effectively when data is standardized

WHAT ABOUT OUR RESEARCHER?

- The data would have been in a standard format and using controlled vocabularies, making integration with their own data seamless.
- Clear documentation & rich metadata would explain variables, methods, and context, avoiding confusion.





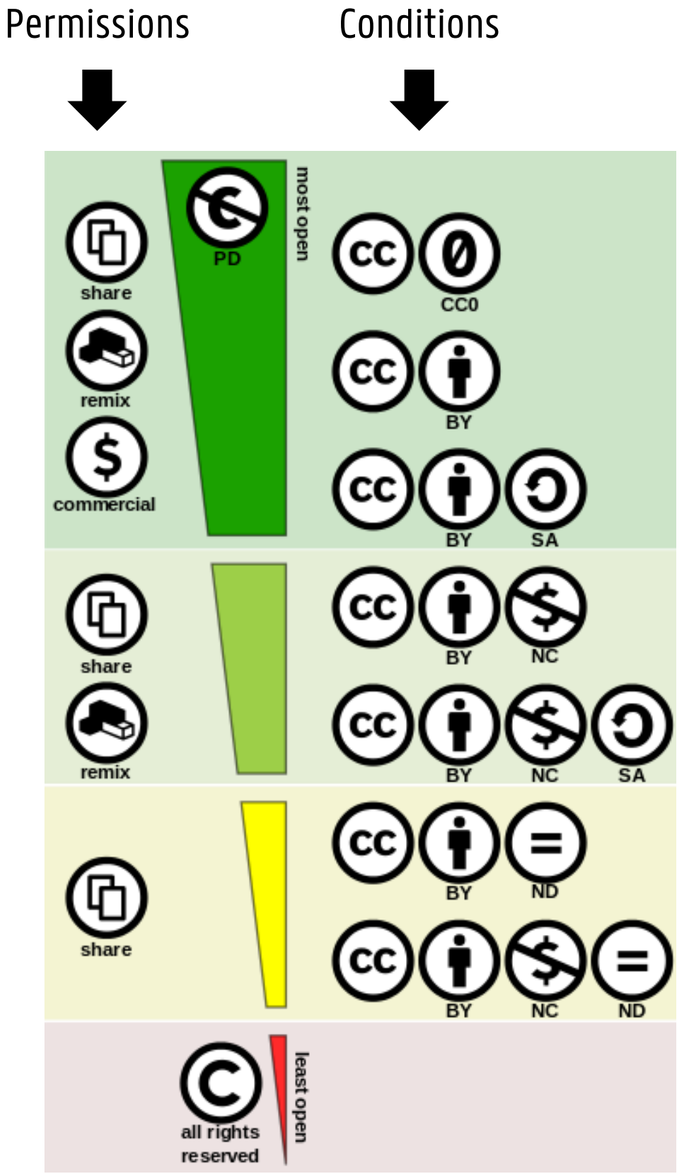
Artwork by Allison Horst

LICENSES

FINDABLE	Persistent identifiers (PIDs)	Rich metadata, includes PID	Machine readable metadata
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INTEROPERABLE	Metadata standards	Vocabularies & ontologies	Linked metadata
REUSABLE	Metadata includes license	Metadata includes provenance	(Meta)data community standards

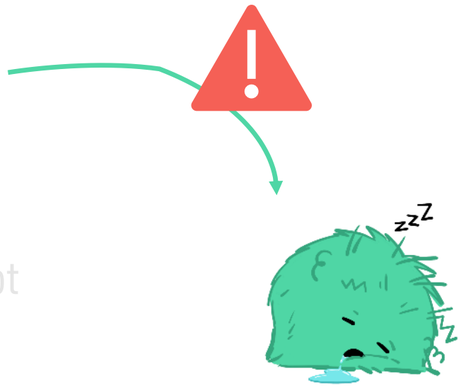
LICENSES

- To give other people the right to **share, use, and build** upon your work.
- Define what can and cannot be done with data in a clear way.
- Licenses are **not automatically assigned**. The owner of rights on protected dataset must attach a license.
- Data ‘publicly available’ without license cannot be accessed, used and shared under copyright or database laws. Licenses allow to **expand rather than restrict use**.
- Use **standard licenses** as much as possible: predefined & simple terms, machine-readable (e.g. Creative Commons, GNU).



WHAT ABOUT OUR RESEARCHER?

- The data would have a license, which would explicitly state what is allowed (e.g., reuse, modification, or redistribution).
- Data could be used immediately (in compliance with the given terms).



More waiting for permission from co-authors…

Artwork by Allison Horst

FAIR ≠ OPEN

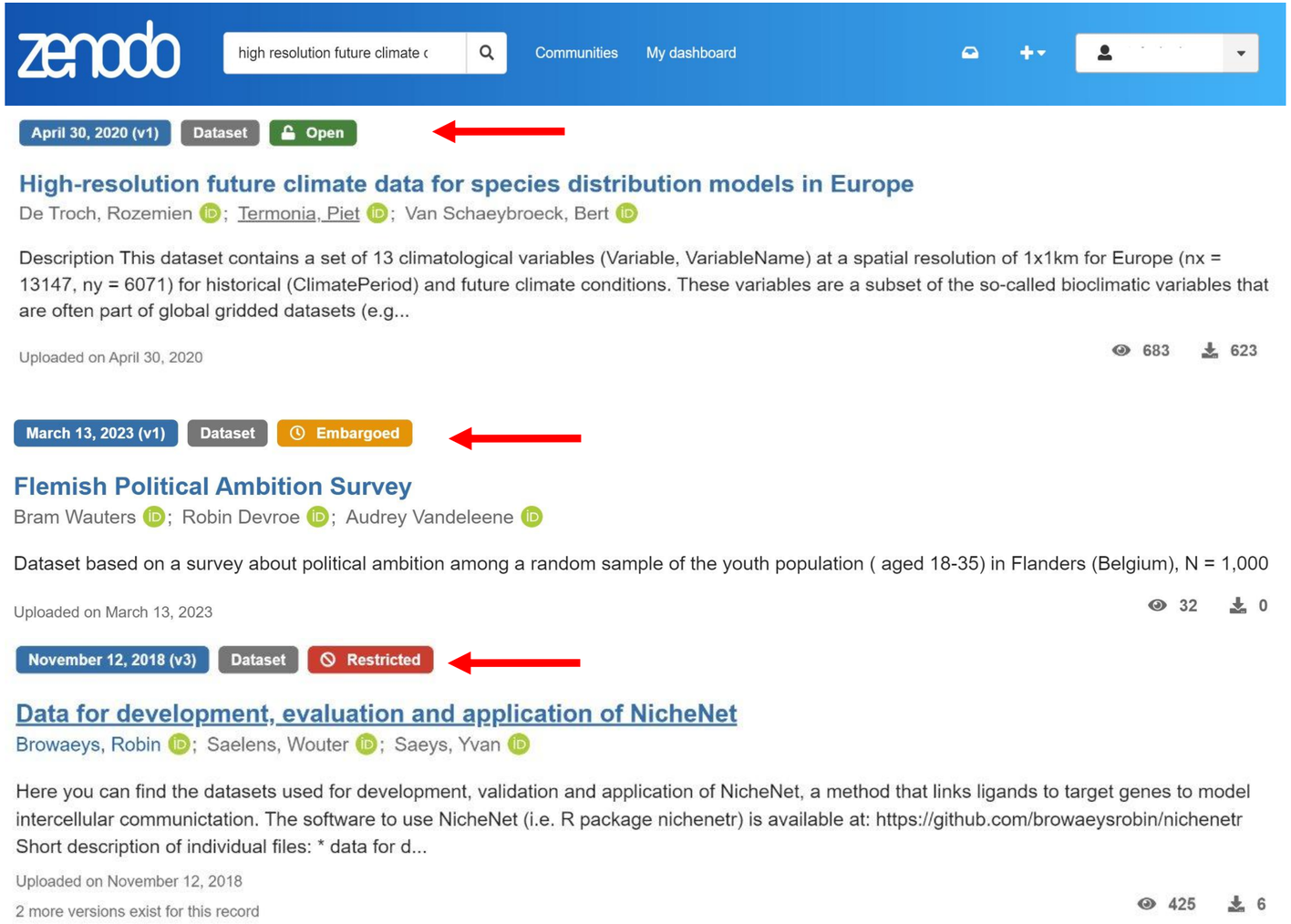
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ACCESSIBLE ≠ OPEN

Sometimes data cannot be (fully) open:

- Personal data (GDPR)
- Third party data
- Contracts/agreements
- Otherwise of confidential / sensitive data

- Metadata should say under which conditions should the data be accessible
  - Authentication & authorization procedures



DATA REPOSITORIES

WAYS OF SHARING DATA

Depositing in a repository explicitly required by some funders and some journal policies

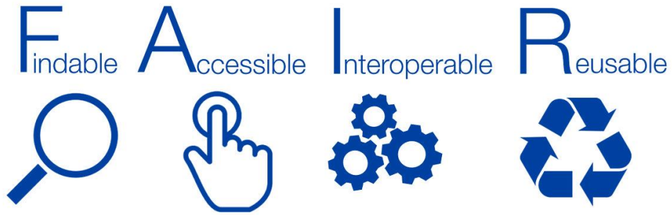
Upon request

Personal or project website

Article suppl. material

Data repository or archive

Research shows that almost non of the data that is requested is actually shared (<11%[1], <7%[2])

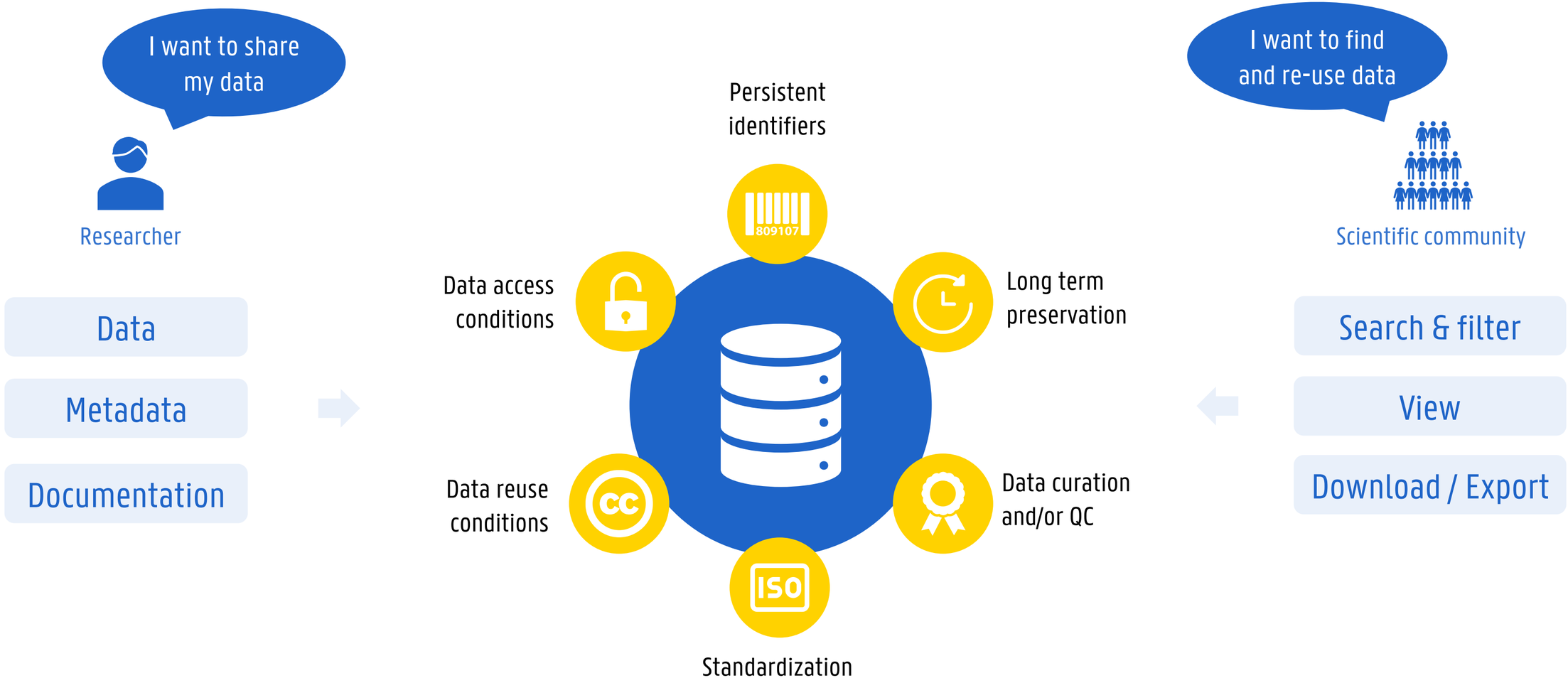


1 Federer, Lisa M et al. (2018) https://doi.org/10.1371/journal.pone.0194768

2 Gabelica et al. (2022), https://doi.org/10.1016/j.jclinepi.2022.05.019

WHAT IS A DATA REPOSITORY?

A database infrastructure that collects, manages and stores data for preservation and sharing



DATA REPOSITORIES & FAIR DATA

→ Key infrastructures that provide functionalities to make your data FAIR.

→ Shared responsibilities between repositories and researchers in making data FAIR

FAIR principle	In practice	Repository	Researcher
FINDABLE	Assign a persistent identifier (PID)	●	
	Data are described with rich metadata	●	●
	Metadata include the PID, are machine readable & searchable online	●	
ACCESSIBLE	(Meta)data retrievable via PID using a standardised communications protocol (e.g. html)	●	
	Authentication and authorisation procedure	●	
INTEROPERABLE	Metadata standards	●	
	Controlled vocabularies	●	●
	References (links) to other (meta)data	●	●
REUSABLE	(Machine readable) license / terms of use	●	●
	Detailed provenance documentation		●
	Domain standards	●	●

Shared responsibilities between repositories and researchers in making data FAIR

TYPES OF DATA REPOSITORIES

GENERIC REPOSITORY

Accepts all kinds of data types, formats, content, disciplines.
Adheres to generic metadata standards

zenodo

Ghent University Community

figshare

DRYAD

DOMAIN-SPECIFIC REPOSITORY

Accepts data from a particular field, discipline, or with a specific format or methodology.
Adheres to domain (meta)data standards

Sodha

Global Biodiversity Information Facility
Free and Open Access to Biodiversity Data

ads ARCHAEOLOGY DATA SERVICE

DataONE
Data Observation Network for Earth

INSTITUTIONAL REPOSITORY

Accepts data that is generated by a specific institution
Hosted by the institution

Edinburgh DataShare

Yoda
research data management

KU LEUVEN RDR

THE FAIR SPECTRUM

- FAIR principles are steps along a continuum toward machine-actionable data.
- Your efforts as a researcher significantly impact the level of FAIRness.
- Different repositories enable varying levels of FAIRness:

Generic repositories
Provide basic FAIR functionalities

Domain-specific repositories
Often score higher due to tailored infrastructure and standards.



MATERIALS DATA FACILITY

Home | Publish + Discovery Guides | How to Cite | Support | Login

Discover Data

Search all MDF and Foundry-ML datasets

tomography | Search

Tags

- [ ] microscopy 14
- [ ] microstructures 8
- [ ] experiment 7
- [ ] machine learning 7
- [x] metals and alloys 7
- [ ] microstructure 4
- [ ] tomography 4
- [ ] Ceramic matrix composite (CMC) 3
- [ ] Polymer derived ceramic (PDC) 3
- [ ] X-ray computed tomography (XCT) 3

Year Published

- [ ] 2019 5
- [ ] 2018 2
- [ ] 2022 2
- [ ] 2016 1
- [ ] 2017 1

XCT dataset for "In-situ 3D visualization of composite microstructure during polymer-to-ceramic conversion"

Larson, Natalie M.; Zok, Frank W.

10.18126/M20355

Preceramic polymer | polymers | Ceramic matrix composite (CMC) | X-ray computed tomography (XCT) | experiment | fibers | microstructure | composites | Polymer derived ceramic (PDC) | tomography | microscopy | Precursor impregnation and pyrolysis (PIP)

Ex-situ XCT dataset for "X-ray computed tomography of microstructure evolution during matrix impregnation and curing in unidirectional fiber beds"

Larson, Natalie M.; Zok, Frank W.

10.18126/M2QM0Z

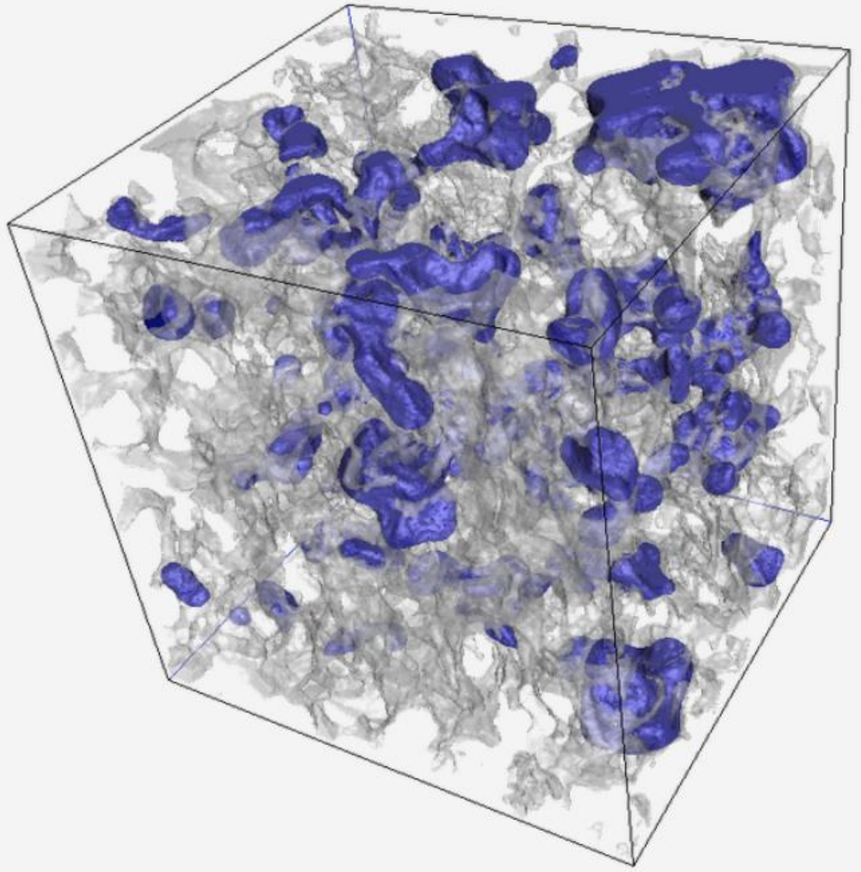
Ceramic matrix composite (CMC) | X-ray computed tomography (XCT) | experimental | Polymer matrix composite (PMC) | microstructure | matrix processing | impregnation | composites | thermodynamics | Polymer derived ceramic (PDC) | curing | microscopy | tomography | fiber rearrangement

In-situ XCT dataset for "Insights from in-situ X-ray computed

Real-time Porosity Mapping and Visualization for Synchrotron

TACC | TEXAS

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Direct simulation of residual phase (disconnected blobs in blue) in Berea Sandstone (imaged based pore grain surface shown in transparent gray).

DIGITAL POROUS MEDIA PORTAL

A data portal for fast storage and retrieval, sharing, organization and analysis of images of varied porous micro-structures

This portal (formerly known as Digital Rocks Portal) enhances research resources for modeling/prediction of porous material properties in the fields of Petroleum, Civil and Environmental Engineering and Geology. It lets users manage, preserve, visualize and analyze available images of porous materials and experiments performed on them, and any accompanying measurements (porosity, capillary pressure, permeability, electrical, NMR and elastic properties, etc.) required for both validation on modeling approaches and the upscaling and building of larger (hydro)geological models.

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KEYWORDS

Authors, clades, epochs, ...

France, dinosaur, ...

SCIENTIFIC DOMAIN

Discipline

- Vertebrate paleontology
- Invertebrate paleontology
- Vertebrate biology
- Paleoanthropology
- Archaeology
- Paleobotany
- Ichnology

GEOLOGICAL TIME

Era

- Mesozoic
- Cenozoic
- Paleozoic
- Neoproterozoic

LOCALISATION

« ‹ Show 20 Page 1 › » 1-20 of 232 results

ibis in jar MG.2038

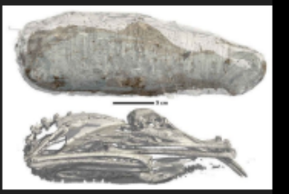
Archaeology Africa Egypt Cenozoic Quaternary



3D reconstruction for ibis in jar MG.2038. MG.2038 is a sealed jar containing an ibis mummy. Its body is relatively well preserved: the feathers are visible, but the soft parts have essentially become dust. The spine is not completely in connection, and some vertebrae and groups of vertebrae are disconnected. Parts of the fabric were also reduced to powder as time passed. Traces on the jar, left by the craftsman's hand, indicate that it was made on a wheel, the container being rotated clockwise. The jar was sealed with a cover glued with a mortar of sorts. The jar also has a hole in its bottom part. This hole's shape and the ceramic fragment found inside the jar could indicate that it is due to striking

ibis MHNGr.ET.1456

Archaeology Africa Egypt Cenozoic Quaternary



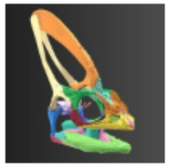
3D reconstruction for ibis MHNGr.ET.1456. MHNGr.ET.1456 is an ibis mummy. The body of the ibis is complete, the feathers are still present, as well as soft parts. Its head is lying on its forelimbs tucked underneath the bird. The mummy was infested with insects, as shown by exit tunnels, the substantial number of pupae, the remains of adult insects and flesh reduced to powder inside the mummy. Many gastropods were discovered inside the body of the ibis, found in the preserved remains of the stomach. A fracture, due to twisting, is visible on one of the cervical vertebrae. It is also possible to notice a pathology affecting the distal part of the two tibiae, which show evidence of osteoporosis. For further details

dog MHNGr.ET.1023

Archaeology Africa Egypt Cenozoic Quaternary



3D reconstruction for dog MHNGr.ET.1023. MHNGr.ET.1023 is a mummy of a dog puppy, relatively not well preserved.



Media 000769445: Skull

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Type Mesh Modality X-Ray Computed Tomography (CT/microCT)

Media represents biological specimen uf:herp:191369 *Chamaeleo calyptratus*

Managed by Jaimi Gray

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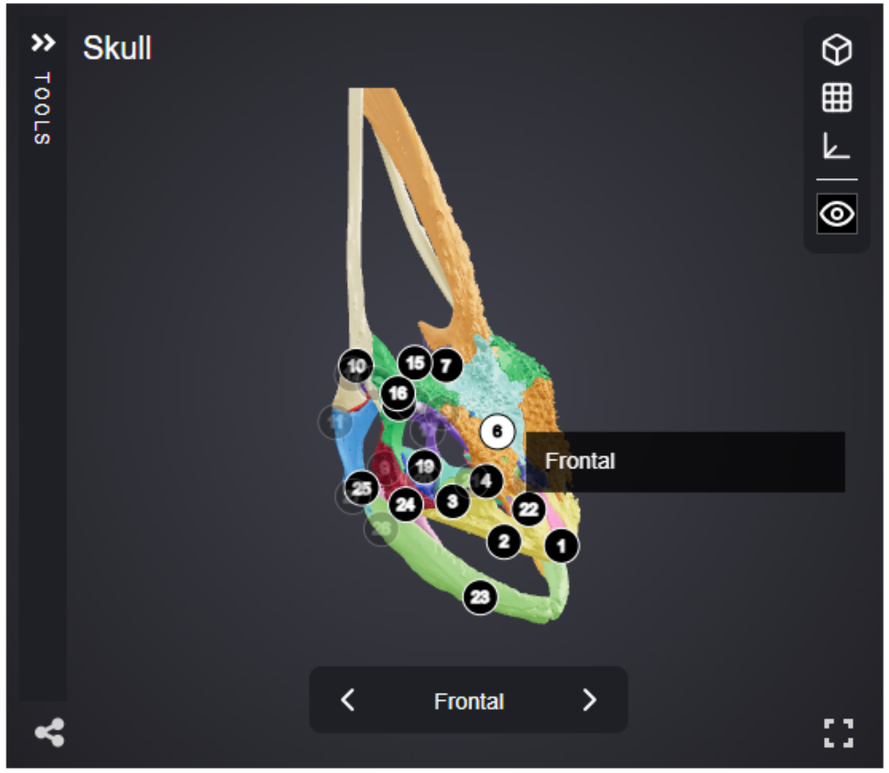
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GENERAL DETAILS

Field Definitions

Media ID	000769445
Media type	Mesh
Object element or part	skull
Object represented	uf:herp:191369
Object taxonomy	*Chamaeleo calyptratus*
Object organization	FLMNH Division of Herpetology
Side	NotApplicable
Orientation	--
Short description	Skull with colored elements
Full description	Skull of a veiled chameleon, Chamaeleo calyptratus, with bones colored and labelled to demonstrate skull anatomy. Digital dissection and scene by Jaimi Gray. 3D model produced from a CT scan of a Florida Museum of Natural History specimen (UF herp 191369). Segmentation and mesh files generated using 3D Slicer. Meshes further edited and animation produced in Blender. Anatomy follows Evans, Susan E. "The skull of lizards and tuatara." Biology of the Reptilia 20 (2008): 1-347. Scanning funded by the oVert TCN project at the University of Florida (NSF DBI-1701714) https://tinyurl.com/oVertwiki
Creator	Jaimi Gray
Date created	August 11, 2025
Date uploaded	August 11, 2025

MEDIA DISPLAY



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Acanthocybium solandri

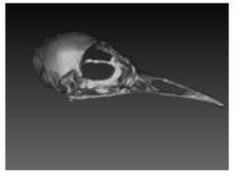

Accipiter nisus

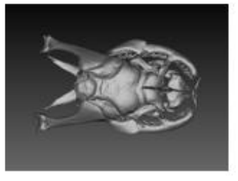

Aceratherium mite


Acontias percivali

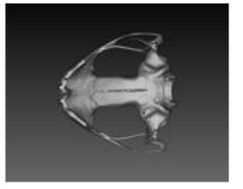

Acontias plumbeus


Acrobates pygmaeus


Acrocephalus arundinaceus


Acrochordus granulatus


Adapis magnus


Adenomera andreae


Adenomus kelaartii

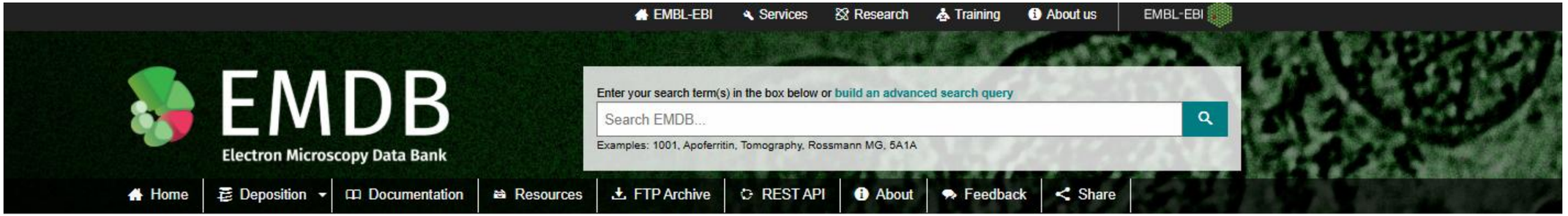








ELECTRON MICROSCOPY DATA BANK



Edit your search query here and hit Return to execute it:

```
database:EMPIAR
```

Filter By

Current Database

- EMPIAR [2265]

Current status

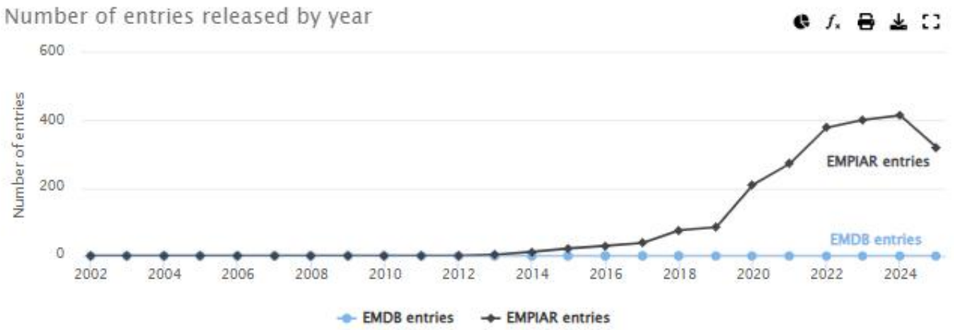
- REL [2265]

Release Date

- 2025 [319]
- 2024 [415]
- 2023 [402]
- 2022 [379]
- 2021 [273]
- 2020 [210]
- 2019 [85]
- 2018 [76]

Author by name

- Potter CS [59]
- Carragher B [58]
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- Shapiro L [45]



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EMPIAR-12856

Related EMDB



Cryo-electron tomography of permeabilised CEM T lymphoblasts incubated with isolated HIV-1 cores

Release date: July 10, 2025 **Experiment type:** EMDB

Dataset size: 9.9 GB

Hou Z

To Be Published

Image sets:

- Motion corrected stacks of tilt series of permeabilised CEM T lymphoblast cells incubated with isolated HIV-1 VLP cores

OPEN NEURO

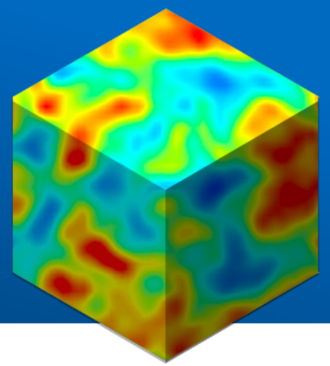
OpenNEURO

SEARCH SUPPORT DOCUMENTATION

OpenNeuro PET

The PET portal of OpenNeuro is supported by a collaboration between Stanford University, NIH, MGH and the Neurobiology Research Unit (NRU) at Copenhagen University Hospital through the OpenNeuroPET project. The project is funded through the BRAIN Initiative and the Novo Nordisk Foundation. Besides developing data sharing, the OpenNeuroPET project also aims at developing user friendly tools for the BIDS based data curation of PET data as well as tools for automated QC and template building.

493 Participants **23** Public Datasets



Search at the participant-level with Neurobagel

Keywords

Enter Keyword(s) to Search

CT ×

Choose Modality

Initiatives

Dataset Type

Age of Participants

Number of Participants

Search PET Portal

These filters return **16** datasets: CLEAR ALL

KEYWORD: CT ×

MODALITY: PET ×

SORT BY: Relevance

Longitudinal (3M, 6M) and cross-sectional (10M, 16M) evaluation of SynVesT-1 PET in zQ175DN mice. ARG and IHC were performed at 16M.

PET

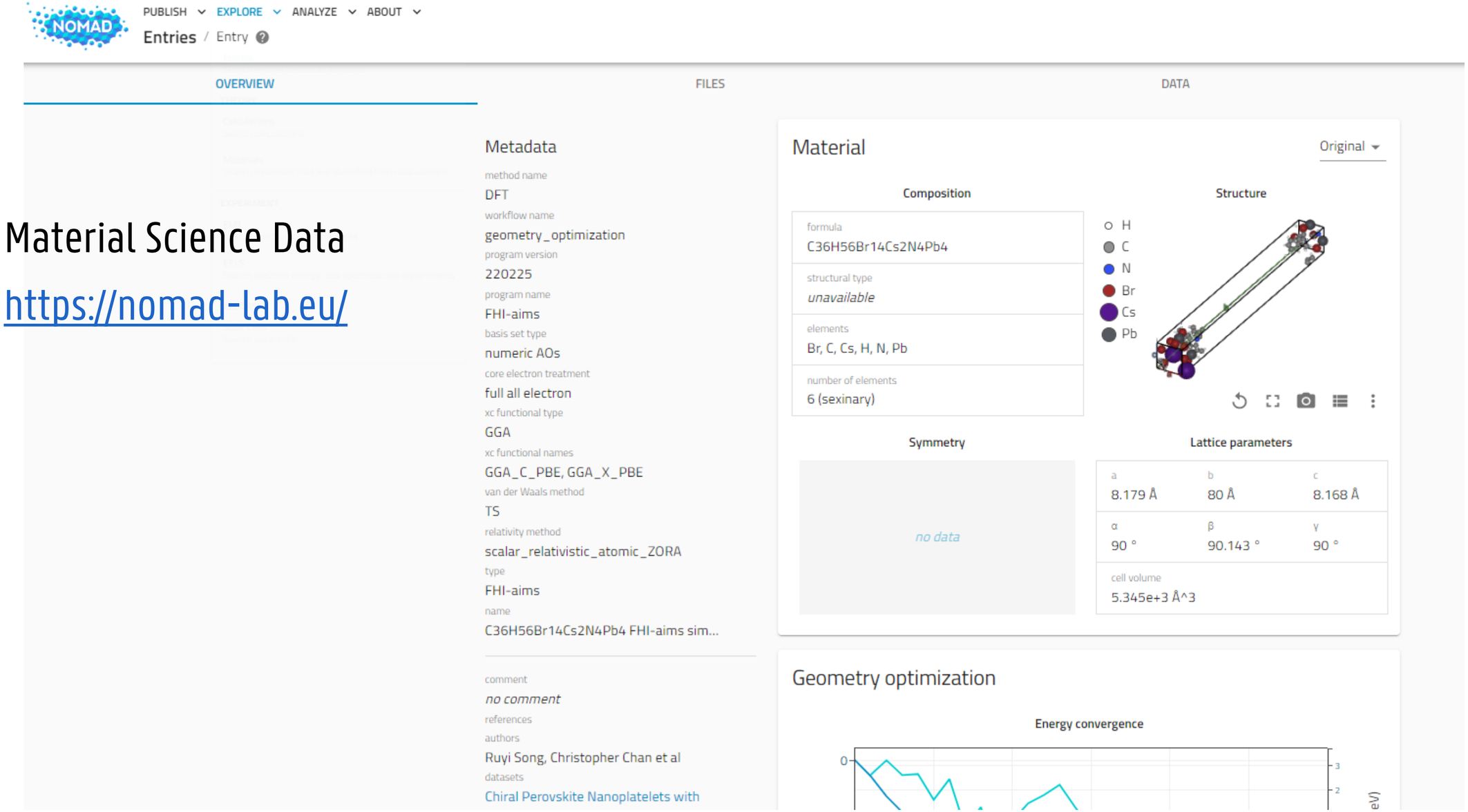
Session 1 - Baseline at 16M, Session 2 - Baseline at 10M, Session 3 - Baseline at 3M, Session 4 - Baseline at 6M. Session 1 and 2 consider cross-sectional cohorts. Session 3 and 4 are longitudinal, meaning some mice but not all mice were scanned at both timepoints.

Show Details

NOMAD

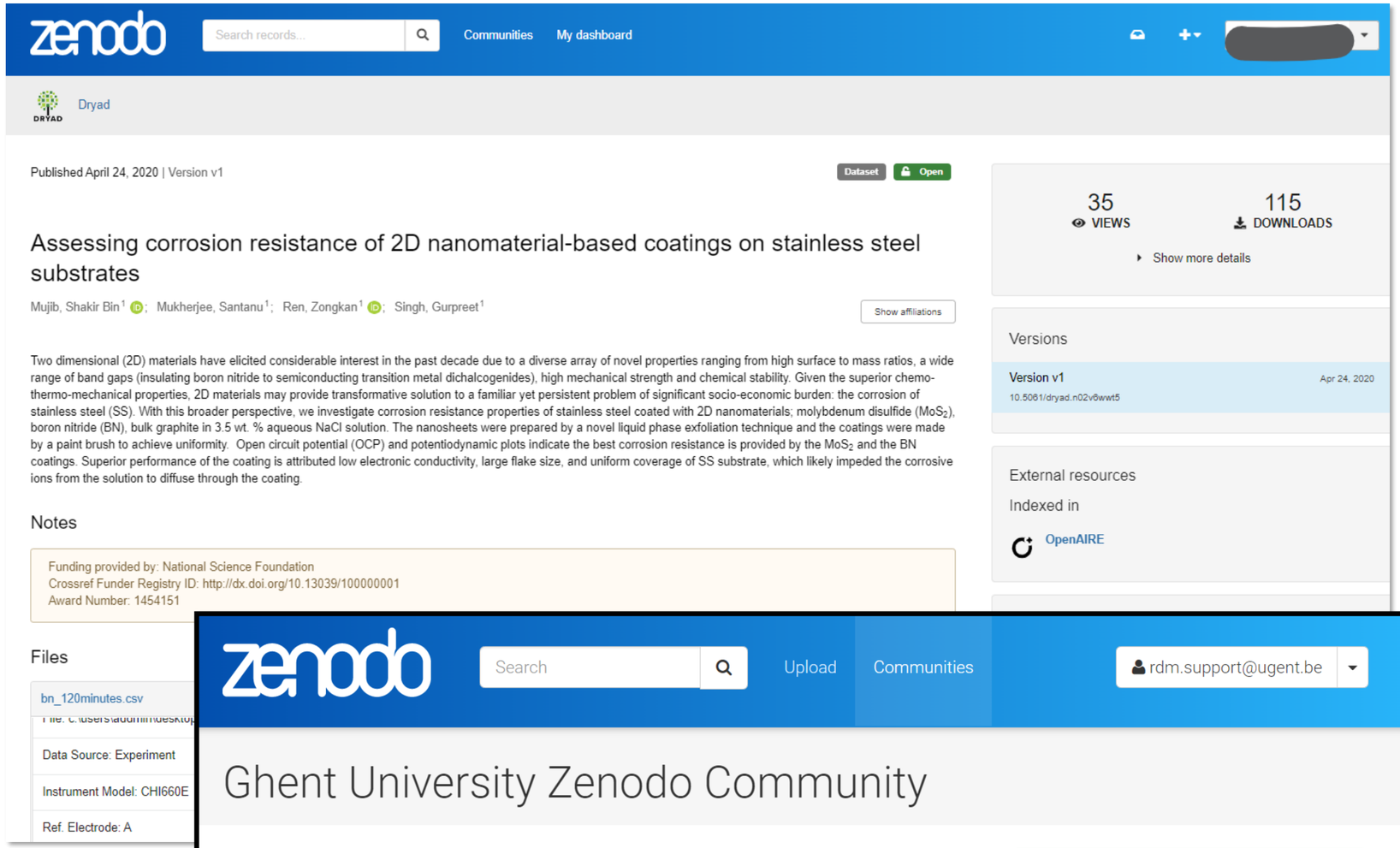
Material Science Data

https://nomad-lab.eu/



GENERAL PURPOSE REPOSITORIES & UGENT ZENODO COMMUNITY

- Base FAIRness level.
- Ugent Zenodo Community
  - Accepts research data and software
  - Ugent data curators
  - Better ranking results
  - Avoid dataset size restrictions
  - Avoid spam filters



B2SHARE

For Communities And Organisations

The B2SHARE option for communities and organizations offers the option for dedicated and branded B2SHARE instances which can be customized to community and organizational needs. This support customization to the metadata schemas, community domains and access rights.

Requested total storage capacity	1TB - higher than 10TB
How long must the data be stored	3 years - more than 10 years

Contact

For Large datasets

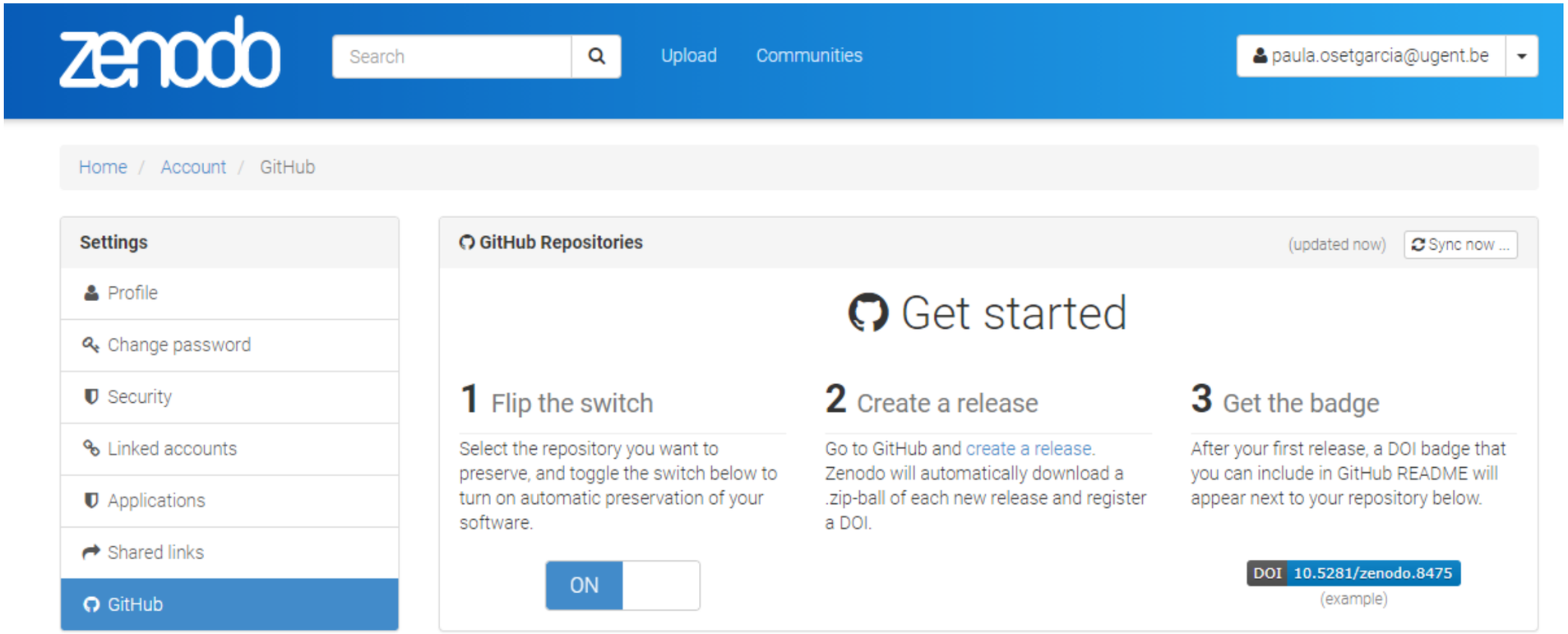
EUDAT is a network service providers across Europe specialist in large ICT infrastructure supporting research. To researchers and/or communities seeking need to publish and/or make large datasets accessible in a FAIR way EUDAT can offer access to data repositories supporting large datasets (>50TB). These repositories are made available through the EUDAT partners.

Requested total storage capacity	50TB - higher than 500TB
How long must the data be stored	3 years - longer than 10 years

Contact

OTHER RESEARCH OUTPUTS
SHARING CODE AND MAKING CODE CITABLE

- GitHub – Zenodo integration: https://guides.github.com/activities/citable-code/



KEY TAKE-AWAYS

- FAIR principles are guidelines designed to improve the **reusability** of data.
- Repositories handle the technical aspects of making data FAIR.
- **FAIR is a spectrum**—it's not either FAIR or not FAIR.
- Discipline-specific repositories often achieve higher FAIR scores due to tailored infrastructure and standards.
- Generalist repositories remain a valuable option for sharing data.
- **FAIR is a shared responsibility**—it also requires effort from you as a researcher.

Paula Oset Garcia

Data Steward

UNIVERSITY SERVICES – RESEARCH

TEAM OPEN SCIENCE

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